

STAT 5700 — Quiz 3**Date:** February 26, 2026**SOLUTIONS**

Suppose you are a nurse working at a blood drive. People with O- (“O-negative”) blood type are coveted, because they are considered “universal donors”, meaning their blood can be donated to people of any blood type. About 8% of the population has type O-. Assume you have 50 donors and are interested in modeling how many are O-. Assume each donor’s blood type is independent of the others’.

- a. What distribution is appropriate to model this scenario? Provide the name of the distribution and the values of its parameter(s).

$$\textit{Binomial}(n = 50, p = 0.08)$$

- b. Write out the probability distribution, $p(y)$, for this scenario. Make sure to include the support.

$$p(y) = \binom{50}{y} (0.08)^y (0.92)^{(50-y)}, \quad y = 0, 1, \dots, 50$$

- c. What is the probability that at least two of the 50 donors are O-?

$$\begin{aligned} P(Y \geq 2) &= 1 - P(Y = 0) - P(Y = 1) \\ &= 1 - \binom{50}{0} (0.08)^0 (0.92)^{50} - \binom{50}{1} (0.08)^1 (0.92)^{49} \\ &= 0.917288 \end{aligned}$$

- d. What is the expected (average) number of O- donors among the 50?

$$E(Y) = np = 50(0.08) = 4$$